INCH-POUND

MIL-DTL-32341A (MR)
w/AMENDMENT 3
21 April 2025
SUPERSEDING
MIL-DTL-32341A (MR)
w/AMENDMENT 2
5 SEPTEMBER 2024

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM, ALLOY 2139 WELDABLE and ALLOY 2195 and 2060 UNWELDABLE APPLIQUE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers three wrought aluminum armor plate alloys for both welded and un-welded applications in nominal thicknesses from 0.500 to 4.000 inch, inclusive (see 6.2). An additional application for the AA2139 alloy with the T84 temper is also included in this specification for ballistic plate with an ordered thickness of 1.500 to 2.600 inches. The weldability of wrought aluminum alloy AA2139 armor has been determined for class I, type A armor for these thicknesses, only; 0.500" and 1.500". The weldability for class I, type B and the other alloys (class II & class III) has not been determined and therefore should only be used as appliqué armor.

1.2 Weldability. Class I material covered by this specification has been demonstrated to be weldable to itself and some other weldable alloys (see 6.3 and 6.4).

1.3 Classification. The wrought aluminum armor should be of the following classes, as specified (see 6.2).

1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 2139 aluminum alloy.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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1.3.1.1 Type A. Type A has a temper designation (See 6.6) which is a high-strength temper designed for maximum resistance to armor piercing (AP) projectiles. The applicable gauge range for class I, type A is 0.500-4.000 inches.

1.3.1.2 Type B. Type B has a temper designation (See 6.6) which is designed for maximum resistance to blast. The applicable gauge range for class I, type B is 1.501 to 2.600 inches.

1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 2195 aluminum alloy. The applicable gauge range for class II is 0.500 - 2.250 inches.

1.3.3 Class III. Class III is wrought aluminum armor that conforms to the Aluminum Association designation for the 2060 aluminum alloy. The applicable gauge range for class III is 0.500 - 2.500 inches.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V_{50} Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

TEST OPERATING PROCEDURES

TOP 2-2-711	-	Proofing Projectile Impact Testing of Armor Weldments

(Copies of these documents are available online at https://quicksearch.dla.mil/).

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2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal should apply.

THE ALUMINUM ASSOCIATION, INC.

Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org/).

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E716	-	Standard Practices for Sampling Aluminum and Aluminum Alloys for Spectrochemical Analysis
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL

AMS 2750	-	Pyrometry
AMS 2772	-	Heat Treatment of Aluminum Alloy Raw Materials

(Copies of these documents are available online at https://www.sae.org/.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical compositions for classes I, II, and III alloys defined by the Aluminum Association Teal Sheets, "International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys" are listed in table I and shall be within those limits. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

TABLE I. Chemical composition, weight percent. 1/

ELEMENTS	SYMBOL	Class I 2139 ALLOY 2/	Class II 2195 ALLOY 2/	Class III 2060 ALLOY 2/
Silicon	Si	0.10	0.12	0.07
Iron	Fe	0.15	0.15	0.07
Copper	Cu	4.5 -5.5	3.7 - 4.3	3.4 - 4.5
Manganese	Mn	0.20 -0.60	0.25	0.10 - 0.50
Magnesium	Mg	0.20 - 0.80	0.25 - 0.8	0.60 - 1.1
Chromium	Cr	0.05	N/A	N/A
Zinc	Zn	0.25	0.25	0.30 - 0.50
Titanium	Ti	0.15	0.10	0.10
Vanadium	V	0.05	N/A	N/A
Zirconium	Zr	N/A	0.08 - 0.16	0.05 - 0.15
Lithium	Li	N/A	0.8 - 1.2	0.60 - 0.90
Silver	Ag	0.15 - 0.60	0.25 - 0.6	0.05 - 0.50
Other, max. Each	---	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

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3.3 Mechanical properties. Unless otherwise specified in the contract or purchase order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal (L) direction shall meet the minimum mechanical properties listed in table IIa for class I or in table IIb for classes II and III. If mechanical property requirements differ from those contained in tables IIa or IIb, or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

TABLE IIa. Minimum mechanical properties. 1/ 2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class I Type A	Class I Type B	Class I Type A	Class I Type B	Class I Type A	Class I Type B
0.500 to 3.000, incl.	67	62 3/	64	53 3/	9	10 3/
3.001 to 4.000, incl.	67	N/A	64	N/A	9	N/A

1/ The test specimen gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
2/ Values are taken in the Longitudinal Direction (see 3.3).
3/ For alloy 2139 (class I, type B) the applicable thickness range is 1.501-2.600 inches.

TABLE IIb. Minimum mechanical properties. 1/ 2/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class II	Class III	Class II	Class III	Class II	Class III
0.500 to 2.500, incl.	71 3/	73	63 3/	70	9 3/	7

1/ The test specimen gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
2/ Values are taken in the Longitudinal Direction (see 3.3).
3/ For alloy 2195 (class II) the applicable thickness range is 0.500-2.250 inches.

3.4 Weldability. When the vehicle requires weldments, the fabricators (OEMs and kit manufacturers, etc.) are required to demonstrate weldability in accordance with MIL-STD-3057, "Arc Welding of Armor Grade Aluminum" for all armor to armor and armor to non-armor applications. This may also require Proofing Projectile Impact Testing (PPIT) for armor to armor weldments in accordance with procedures in TOP 2-2-711, "Proofing Projectile Impact Testing of Armor Weldments" (revision A and above) depending on the weld classification in accordance with MIL-STD-3057. PPIT procedures and associated striking velocities for established welded class I, class II, and class III plate thicknesses are listed in TOP 2-2-711. For any newly introduced armor material, configuration, or unproven thickness being used for armor applications, the aluminum fabricators shall qualify a weld procedure in accordance with MIL-STD-3057, and this requires PPIT testing following TOP 2-2-711. Since ordered thicknesses of the alloys referenced

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by this specification may not be listed in TOP 2-2-711, it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3.

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.6 Stress corrosion resistance. Stress Corrosion Cracking resistance is an optional requirement under this specification. When stress corrosion testing is requested in the contract or purchase order (see 6.2) and unless otherwise specified in the contract or purchase order (see 6.2), plate 0.750 inch and over in thickness shall be resistant to stress corrosion cracking, with a minimum of 5 of the 9 specimens showing no evidence of cracking at the end of 96 hours when tested as specified in 4.7.5.

3.7 Thermal processing. Heat treatment shall conform to the requirements of SAE AMS 2750 and SAE AMS 2772 and shall be such as to enable the material to meet the requirements of these specifications.

3.8 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or purchase order (see 6.2).

3.8.1 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2, except for thickness.

3.8.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in table III.

TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)							
	OVER	0.00	39.37	59.06	78.74	98.43	118.11	137.80	157.48
	THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)							
0.500	1.000	0.031	0.031	0.037	0.043	0.051	0.060	0.070	0.085
1.000	1.575	0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362	0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.000	0.075	0.075	0.085	0.100	0.105	0.125	---	---
3.000	4.000	0.090	0.090	0.105	0.110	0.125	---	---	---

3.9 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8

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of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.9.1 Ballistic test plates. In addition to the markings in 3.9, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.10 Ballistic test plate information. For each lot of aluminum alloy armor a properly completed Aluminum Armor Test Data Form (See figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.11 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use. Minor scratches, gouges, dents and similar damage (see 3.11.2) are acceptable if the thickness is not reduced below the minimum material tolerance thickness.

3.11.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

3.11.2 Imperfections. The depth of surface scratches, nicks, pits, or gouges shall not reduce the plate thickness below the allowable minimum thickness. Isolated individual pits over 0.015 inches deep but not over 0.030 inches deep and not within 6 inches of each other and do not violate the minimum allowable thickness, are acceptable.

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR										
SPECIFICATION: MIL-DTL-32341			**REVISION: A**			**AMENDMENT: 3**				
FIRING RECORD:		**DATE:**			**TEST CONTRACT NO:**					
	PLATE MANUFACTURER / PRODUCER				**PRIME CONTRACTOR**					
Name:										
Address:										
POC:										
Phone No:										
Email:										
TEST ITEM IDENTIFICATION:					Ordered Thickness:					
Lot No. _____ **Plate No.** _____			**Alloy:** _____ **Temper** _____				**Class:** _____ **Type:** _____			
PURPOSE: ___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing										
SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)										
CHEMISTRY ANALYSIS	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**	
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**			**Other - Total**	**Al**
										Rem.
MECHANICAL PROPERTIES:										
UTS (ksi):			0.2% YS (ksi):				Elongation (%):			
WELDABILITY:	Thickness		Required Velocity		Actual Velocity		Pass/Fail			
BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES ____ NO ____**										
	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**					
Test:										
Projectile:										
	NOTES:									
SPALL:	Spall Size:		Remarks:							
DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT & SUPPLIER OFFICIALS										
LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-32341										
Government Representative:		Date:	Supplier Representative:			Date:				

FIGURE 1. Aluminum Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all plate(s) of the same alloy and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate(s) in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit. All lots shall be fully traceable back to the ingot source by the supplier.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use on production.

4.3.1 First article inspection. First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examinations of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2).

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4.5.1.4 Stress corrosion tests. When required (see 6.2), nine (9) stress corrosion test specimens shall be removed from the same plate that has been selected for ballistic testing. The location and type of specimen shall be as specified in 4.7.5.

4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.4.

4.5.2.4 Stress corrosion tests. When required (see 6.2), nine test specimens shall be prepared from one plate randomly selected from each lot for stress corrosion testing. The plate shall meet the requirements when tested as specified in 4.7.5.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.9) and workmanship (see 3.11) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.8 in accordance with the sampling procedures approved by the

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procuring activity and as specified in the contract or purchase order (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.

4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251, and E3061. In case of dispute, analysis by method E3061 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal (L) direction. For plate less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Weldability. When required (see 6.2), the thickness of the weldments specified in the contract or drawing shall be used to determine the proofing projectile and striking velocity in accordance with MIL-STD-3057 and TOP 2-2-711.

4.7.4 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL(P) for that plate.

4.7.4.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

4.7.4.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

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TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	TABLE	CLASS I (2139)	CLASS II (2195)	CLASS III (2060)
0.500 - 0.749	Cal. .30 AP M2	30	A-I	LISTED	LISTED	LISTED
0.750 - 0.950	Cal. .50 FSP	0	A-II	LISTED - SAME V_{50}		LISTED
0.951 - 1.500	20-mm FSP	0	A-III	LISTED - SAME V_{50}		
0.750 - 1.500	Cal. .30 AP M2	0	A-IV	LISTED - SAME V_{50}		LISTED
1.501 - 2.600	Cal. .50 AP M2	0	A-V	LISTED 1/	LISTED 2/	LISTED 3/
2.501 - 3.500	14.5-mm BS41	0	A-VI	LISTED	N/A 2/	N/A 3/
3.501 – 4.000	20-mm M602	0	A-VII	TBD	N/A 2/	N/A 3/

1/ Class I, type A (alloy 2139) the applicable gauge for ordered thickness ends at 2.500 inches.
2/ Class II (alloy 2195) the applicable gauge for ordered thickness ends at 2.250 inches.
3/ Class III (alloy 2060) the applicable gauge for ordered thickness ends at 2.500 inches.

4.7.5 Stress corrosion specimens. When stress corrosion testing is required (see 6.2), Short transverse stress corrosion test specimens shall be prepared in accordance with ASTM G38 and tested in accordance with ASTM G47. Nine specimens shall be tested per sample per lot. The specimens shall be stressed to 45 ksi. A maximum delay of 3 hours between stressing and initiation of stress corrosion test is permitted.

4.8 Rejection and retest. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2 and 4.8.3.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails, the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic retest. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.8.3 Stress corrosion retest. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense submit thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.5. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.4 and 4.7.5.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing the manufacturer agrees to maintain the same manufacturing procedures. Testing for a

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given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat and tactical vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Pratt & Miller Engineering who has been developing the OCP-TECD CAMEL demonstrator plans to use a weldable 2139 aluminum alloy.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3)
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties and ballistic requirements, if required (see 3.3).
(e) When stress corrosion testing is required and if the test conditions are different (see 3.6, 4.5.1.4, 4.5.2.4, and 4.7.5).
(f) Dimension and tolerance requirements if different (see 3.8 and 3.8.1).
(g) If markings are different and when impression stamping is permissible (see 3.9).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.11.1).
(i) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(j) The orientation of the ballistic plate with respect to rolling is different (see 4.5.1.3 and 4.5.2.3).
(k) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(l) If approval was requested and received for a different ballistic testing facility (see 4.7.3.1).
(m) Rejection and retest requirement, if different (see 4.8).
(n) If reduced testing is allowed (see 4.9).

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(o) Packaging requirements (see 5.1).

(p) Striking velocities for those thicknesses not covered (see MIL-STD-3057 and TOP 2-2-711).

6.3 Weldability. The Army Research Lab has conducted testing and found that class I, type A of AA2139 can be welded and demonstrate good shock resistance when prepared and tested in accordance with MIL-STD-3057 and TOP 2-2-711. If these alloys in thicknesses required for manufacturing are not currently listed in TOP 2-2-711, the fabricator shall follow the procedures of TOP 2-2-711 to generate the required striking velocities before the Proofing Projectile Impact Testing (PPIT) is performed. The fabricator performing welding on these alloys for armor applications is responsible for developing and qualifying welding procedures in accordance with MIL-STD-3057 and TOP 2-2-711. Tests have demonstrated that material covered by this specification is weldable to itself and other weldable 2000 series alloys using a 2000 or 4000 series filler and is weldable to 6000 series alloys using a 4000 series filler. It is not weldable to 5000 or 7000 series alloys by conventional welding techniques. Any deviations from these conditions should be demonstrated in accordance with MIL-STD-3057 prior to acceptance by the procuring activity.

6.4 Weld wire. Corner joints were successfully welded with AA4043 weld wire for class I, type A (AA2139) material when tested with the required proofing projectile per MIL-STD-3057 (see 1.2).

6.5 Density. The density of the specified alloys listed in this specification are as follows:

AA2139 is 0.102 lb/in^3 (2.81 g/ cm^3)
AA2195 is 0.098 lb/in^3 (2.71 g/cm^3)
AA2060 is 0.098lb/in^3 (2.72 g/cm^3)

6.6 Tempers. The following tempers have shown to produce acceptable material when used to obtain the desired condition; maximum resistance to armor piercing (AP) projectiles, or maximum resistance to blast.

For class I, type A = T8
For class I, type B = T84

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6.7 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.8 Subject term (key word) listing.

Ballistic testing	Caliber .50 FSP	20-mm FSP
Caliber .30 AP M2	Military vehicles	20-mm API-T M602
Caliber .50 AP M2	Stress corrosion	14.5-mm API

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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APPENDIX A

ARMOR PLATE, ALUMINUM, ALLOY 2139 WELDABLE & ALLOY 2195 and 2060 UNWELDABLE APPLIQUE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, AA2139, AA2195, and AA2060 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification, and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any fair impact which is not a complete penetration may be considered a partial penetration.

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APPENDIX A

A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-VII.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P)

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shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at their own expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at their own expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
0.415	1311	1349	1440	0.525	1572	1608	1692
0.420	1324	1362	1452	0.530	1583	1619	1702
0.425	1336	1374	1465	0.535	1594	1630	1713
0.430	1349	1387	1477	0.540	1605	1640	1723
0.435	1362	1399	1489	0.545	1615	1651	1734
0.440	1374	1412	1501	0.550	1626	1661	1744
0.445	1387	1424	1513	0.555	1636	1672	1754
0.450	1399	1436	1525	0.560	1647	1682	1764
0.455	1411	1448	1536	0.565	1657	1692	1775
0.460	1423	1460	1548	0.570	1667	1703	1785
0.465	1435	1472	1560	0.575	1678	1713	1795
0.470	1447	1484	1571	0.580	1688	1723	1805
0.475	1459	1496	1582	0.585	1698	1733	1815
0.480	1471	1508	1594	0.590	1708	1743	1824
0.485	1482	1519	1605	0.595	1718	1753	1834
0.490	1494	1531	1616	0.600	1728	1763	1844
0.495	1505	1542	1627	0.605	1738	1773	1854
0.500 1/	1517	1553	1638	0.610	1748	1783	1863
0.505	1528	1564	1649	0.615	1757	1792	1873
0.510	1539	1575	1660	0.620	1767	1802	1882
0.515	1550	1586	1670	0.625	1777	1812	1892
0.520	1561	1597	1681	0.630	1787	1821	1901

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity (continued).

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
0.635	1796	1831	1911	0.740	1986	2021	2099
0.640	1806	1840	1920	0.745	1995	2030	2107
0.645	1815	1850	1929	0.749 2/	2002	2037	2114
0.650	1824	1859	1939	0.750	2004	2038	2116
0.655	1834	1869	1948	0.755	2012	2047	2124
0.660	1843	1878	1957	0.760	2021	2055	2132
0.665	1852	1887	1966	0.765	2029	2064	2141
0.670	1862	1896	1975	0.770	2038	2072	2149
0.675	1871	1905	1984	0.775	2046	2081	2157
0.680	1880	1915	1993	0.780	2054	2089	2166
0.685	1889	1924	2002	0.785	2063	2097	2174
0.690	1898	1933	2011	0.790	2071	2105	2182
0.695	1907	1942	2020	0.795	2079	2114	2190
0.700	1916	1951	2029	0.800	2087	2122	2199
0.705	1925	1960	2038	0.805	2096	2130	2207
0.710	1934	1969	2047	0.810	2104	2138	2215
0.715	1943	1977	2055	0.815	2112	2146	2223
0.720	1952	1986	2064	0.820	2120	2155	2231
0.725	1960	1995	2073	0.825	2128	2163	2239
0.730	1969	2004	2081	0.830	2136	2171	2247
0.735	1978	2012	2090	0.835	2144	2179	2255

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Required BL(P), fps
	Class I & II	Class III		Class I & II	Class III
0.665	1736	1808	0.765	2136	2200
0.670	1754	1826	0.770	2158	2222
0.675	1772	1844	0.775	2181	2243
0.680	1791	1862	0.780	2203	2266
0.685	1809	1881	0.785	2226	2288
0.690	1828	1899	0.790	2250	2310
0.695	1847	1918	0.795	2273	2333
0.700	1866	1937	0.800	2297	2356
0.705	1886	1956	0.805	2321	2379
0.710	1906	1975	0.810	2345	2403
0.715	1925	1995	0.815	2369	2426
0.720	1946	2014	0.820	2394	2450
0.725	1966	2034	0.825	2419	2474
0.730	1986	2054	0.830	2444	2499
0.735	2007	2074	0.835	2470	2523
0.740	2028	2095	0.840	2495	2548
0.745	2049	2115	0.845	2521	2573
0.750 1/	2070	2136	0.850	2548	2599
0.755	2092	2157	0.855	2574	2624
0.760	2114	2178	0.860	2601	2650

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps Class I & II	Required BL(P), fps Class III	Thickness, inches	Required BL(P), fps Class I & II	Required BL(P), fps Class III
0.865	2628	2676	0.955	3168	3192
0.870	2656	2703	0.960	3201	3224
0.875	2683	2729	0.965	3234	3256
0.880	2711	2756	0.970	3268	3288
0.885	2739	2783	0.975	3302	3320
0.890	2768	2811	0.980	3336	3353
0.895	2797	2838	0.985	3371	3386
0.900	2826	2866	0.990	3406	3419
0.905	2856	2894	0.995	3442	3453
0.910	2885	2923	1.000	3477	3487
0.915	2915	2952	1.005	3514	3521
0.920	2946	2981	1.010	3550	3556
0.925	2976	3010	1.015	3587	3591
0.930	3007	3040	1.020	3625	3626
0.935	3039	3070	1.025	3662	3662
0.940	3071	3100	1.030	3701	3698
0.945	3103	3131	1.035	3739	3734
0.950 2/	3135	3161	1.040	---	---

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Classes I, II & III		Classes I, II & III
0.865	1171	0.990	1391
0.870	1179	0.995	1401
0.875	1187	1.000	1412
0.880	1195	1.005	1420
0.885	1203	1.010	1430
0.890	1212	1.015	1440
0.895	1220	1.020	1450
0.900	1228	1.025	1460
0.905	1237	1.030	1470
0.910	1246	1.035	1480
0.915	1254	1.040	1490
0.920	1263	1.045	1501
0.925	1272	1.050	1511
0.930	1280	1.055	1522
0.935	1289	1.060	1532
0.940	1298	1.065	1543
0.945	1307	1.070	1553
0.950	1316	1.075	1564
0.951 1/	1318	1.080	1575
0.955	1325	1.085	1586
0.960	1335	1.090	1597
0.965	1344	1.095	1608
0.970	1353	1.100	1619
0.975	1362	1.105	1630
0.980	1372	1.110	1642
0.985	1381	1.115	1653

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20mm fragment simulating projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps Classes I, II & III	Thickness, inches	Required BL(P), fps Classes I, II & III
1.120	1664	1.245	1978
1.125	1676	1.250	1991
1.130	1687	1.255	2005
1.135	1699	1.260	2019
1.140	1711	1.265	2033
1.145	1723	1.270	2047
1.150	1735	1.275	2061
1.155	1747	1.280	2076
1.160	1759	1.285	2090
1.165	1771	1.290	2104
1.170	1783	1.295	2119
1.175	1796	1.300	2134
1.180	1808	1.305	2148
1.185	1821	1.310	2163
1.190	1833	1.315	2178
1.195	1846	1.320	2193
1.200	1859	1.325	2209
1.205	1872	1.330	2224
1.210	1884	1.335	2239
1.215	1898	1.340	2255
1.220	1911	1.345	2270
1.225	1924	1.350	2286
1.230	1937	1.355	2302
1.235	1951	1.360	2318
1.240	1964	1.365	2334

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20mm fragment simulating projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps Classes I, II & III	Thickness, inches	Required BL(P), fps Classes I, II & III
1.370	2350	1.490	2773
1.375	2366	1.495	2793
1.380	2383	1.500 2/	2812
1.385	2399	1.505	2831
1.390	2416	1.510	2851
1.395	2433	1.515	2871
1.400	2449	1.520	2891
1.405	2466	1.525	2911
1.410	2484	1.530	2931
1.415	2501	1.535	2951
1.420	2518	1.540	2972
1.425	2535	1.545	2992
1.430	2553	1.550	3013
1.435	2571	1.555	3034
1.440	2589	1.560	3055
1.445	2606	1.565	3076
1.450	2624	1.570	3097
1.455	2643	1.575	3119
1.460	2661	1.580	3140
1.465	2679	1.585	3162
1.470	2698	1.590	3184
1.475	2717	1.595	3206
1.480	2735	1.600	3228
1.485	2754	1.605	3251

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I & II	Class III		Class I & II	Class III
0.665	1693	1711	0.830	1952	1995
0.670	1701	1720	0.835	1959	2003
0.675	1710	1729	0.840	1966	2011
0.680	1718	1738	0.845	1974	2019
0.685	1726	1748	0.850	1981	2027
0.690	1735	1757	0.855	1988	2035
0.695	1743	1766	0.860	1995	2043
0.700	1751	1775	0.865	2003	2051
0.705	1759	1784	0.870	2010	2059
0.710	1767	1793	0.875	2017	2066
0.715	1775	1802	0.880	2024	2074
0.720	1783	1811	0.885	2031	2082
0.725	1791	1819	0.890	2038	2089
0.730	1799	1828	0.895	2045	2097
0.735	1807	1837	0.900	2052	2105
0.740	1815	1846	0.905	2059	2112
0.745	1823	1854	0.910	2066	2120
0.750 1/	1831	1863	0.915	2073	2127
0.755	1839	1871	0.920	2080	2135
0.760	1846	1880	0.925	2086	2142
0.765	1854	1888	0.930	2093	2150
0.770	1862	1897	0.935	2100	2157
0.775	1870	1905	0.940	2107	2165
0.780	1877	1914	0.945	2114	2172
0.785	1885	1922	0.950	2120	2179
0.790	1892	1930	0.955	2127	2187
0.795	1900	1939	0.960	2134	2194
0.800	1907	1947	0.965	2141	2201
0.805	1915	1955	0.970	2147	2209
0.810	1922	1963	0.975	2154	2216
0.815	1930	1971	0.980	2161	2223
0.820	1937	1979	0.985	2167	2230
0.825	1945	1987	0.990	2174	2237

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I & II	Class III		Class I & II	Class III
0.995	2180	2245	1.160	2387	2469
1.000	2187	2252	1.165	2393	2475
1.005	2193	2259	1.170	2399	2482
1.010	2200	2266	1.175	2405	2488
1.015	2206	2273	1.180	2411	2494
1.020	2213	2280	1.185	2417	2501
1.025	2219	2287	1.190	2423	2507
1.030	2226	2294	1.195	2429	2514
1.035	2232	2301	1.200	2435	2520
1.040	2239	2308	1.205	2440	2526
1.045	2245	2315	1.210	2446	2533
1.050	2251	2322	1.215	2452	2539
1.055	2258	2329	1.230	2470	2558
1.060	2264	2335	1.220	2458	2545
1.065	2270	2342	1.225	2464	2551
1.070	2277	2349	1.235	2475	2564
1.075	2283	2356	1.240	2481	2570
1.080	2289	2363	1.245	2487	2576
1.085	2295	2369	1.250	2493	2583
1.090	2302	2376	1.255	2498	2589
1.095	2308	2383	1.260	2504	2595
1.100	2314	2390	1.265	2510	2601
1.105	2320	2396	1.270	2515	2607
1.110	2326	2403	1.275	2521	2613
1.115	2332	2410	1.280	2527	2620
1.120	2339	2416	1.285	2532	2626
1.125	2345	2423	1.290	2538	2632
1.130	2351	2429	1.295	2544	2638
1.135	2357	2436	1.300	2549	2644
1.140	2363	2443	1.305	2555	2650
1.145	2369	2449	1.310	2561	2656
1.150	2375	2456	1.315	2566	2662
1.155	2381	2462	1.320	2572	2668

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I & II	Class III		Class I & II	Class III
1.325	2577	2674	1.445	2707	2814
1.330	2583	2680	1.450	2712	2820
1.335	2588	2686	1.455	2718	2825
1.340	2594	2692	1.460	2723	2831
1.345	2599	2698	1.465	2728	2837
1.350	2605	2704	1.470	2733	2842
1.355	2610	2710	1.475	2739	2848
1.360	2616	2716	1.480	2744	2853
1.365	2621	2721	1.485	2749	2859
1.370	2627	2727	1.490	2754	2865
1.375	2632	2733	1.495	2759	2870
1.335	2588	2686	1.500 2/	2765	2876
1.340	2594	2692	1.505	2770	2881
1.345	2599	2698	1.510	2775	2887
1.350	2605	2704	1.515	2780	2892
1.355	2610	2710	1.520	2785	2898
1.360	2616	2716	1.525	2790	2903
1.365	2621	2721	1.530	2796	2909
1.370	2627	2727	1.535	2801	2914
1.375	2632	2733	1.540	2806	2920
1.380	2638	2739	1.545	2811	2925
1.385	2643	2745	1.550	2816	2931
1.390	2648	2751	1.555	2821	2936
1.395	2654	2756	1.560	2826	2942
1.400	2659	2762	1.565	2831	2947
1.405	2665	2768	1.570	2836	2953
1.410	2670	2774	1.575	2841	2958
1.415	2675	2780	1.580	2846	2963
1.420	2681	2785	1.585	2851	2969
1.425	2686	2791	1.590	2856	2974
1.430	2691	2797	1.595	2861	2980
1.435	2697	2802	1.600	2866	2985
1.440	2702	2808	1.605	2871	2990

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
1.480	2070	2083	2066	1.630	2193	2210	2193
1.485	2074	2087	2070	1.635	2197	2214	2197
1.490	2079	2092	2075	1.640	2201	2219	2201
1.495	2083	2096	2079	1.645	2205	2223	2205
1.501 1/,2/	2088	2100	2084	1.650	2209	2227	2209
1.505	2091	2105	2088	1.655	2213	2231	2213
1.510	2095	2109	2092	1.660	2217	2235	2217
1.515	2100	2113	2096	1.665	2221	2239	2221
1.520	2104	2118	2100	1.670	2225	2243	2225
1.525	2108	2122	2105	1.675	2229	2247	2229
1.530	2112	2126	2109	1.680	2233	2251	2233
1.535	2116	2131	2113	1.685	2237	2255	2237
1.540	2120	2135	2118	1.690	2241	2259	2241
1.545	2124	2139	2122	1.695	2245	2263	2245
1.550	2129	2143	2126	1.700	2248	2267	2249
1.555	2133	2148	2130	1.705	2252	2271	2253
1.560	2137	2152	2134	1.710	2256	2275	2257
1.565	2141	2156	2139	1.715	2260	2279	2261
1.570	2145	2160	2143	1.720	2264	2283	2265
1.575	2149	2165	2147	1.725	2268	2287	2269
1.580	2153	2169	2151	1.705	2252	2271	2253
1.585	2157	2173	2155	1.710	2256	2275	2257
1.590	2161	2177	2160	1.715	2260	2279	2261
1.595	2165	2181	2164	1.720	2264	2283	2265
1.600	2169	2185	2168	1.725	2268	2287	2269
1.605	2173	2190	2172	1.730	2272	2291	2273
1.610	2177	2194	2176	1.735	2276	2295	2277
1.615	2181	2198	2180	1.740	2279	2299	2281
1.620	2185	2202	2184	1.745	2283	2303	2285
1.625	2189	2206	2189	1.750	2287	2307	2289

1/ Specification requirements begin for this ordered thickness.
2/ Specification class I, type B minimum required ballistic limits shall be -90fps vs. listed thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
1.755	2291	2311	2293	1.880	2384	2408	2389
1.760	2295	2315	2297	1.885	2388	2412	2393
1.765	2298	2319	2301	1.890	2392	2415	2397
1.770	2302	2323	2305	1.895	2395	2419	2400
1.775	2306	2327	2309	1.900	2399	2423	2404
1.780	2310	2331	2313	1.905	2403	2427	2408
1.785	2314	2335	2316	1.910	2406	2431	2412
1.790	2317	2339	2320	1.915	2410	2434	2415
1.795	2321	2343	2324	1.920	2414	2438	2419
1.800	2325	2346	2328	1.925	2417	2442	2423
1.805	2329	2350	2332	1.930	2421	2446	2427
1.810	2332	2354	2336	1.935	2424	2449	2430
1.815	2336	2358	2340	1.940	2428	2453	2434
1.820	2340	2362	2344	1.945	2432	2457	2438
1.825	2344	2366	2347	1.950	2435	2460	2441
1.830	2347	2370	2351	1.955	2439	2464	2445
1.835	2351	2374	2355	1.960	2442	2468	2449
1.840	2355	2377	2359	1.965	2446	2472	2452
1.845	2359	2381	2363	1.970	2450	2475	2456
1.850	2362	2385	2366	1.975	2453	2479	2460
1.830	2347	2370	2351	1.980	2457	2483	2463
1.835	2351	2374	2355	1.985	2460	2486	2467
1.840	2355	2377	2359	1.990	2464	2490	2471
1.845	2359	2381	2363	1.995	2467	2494	2474
1.850	2362	2385	2366	2.000	2471	2497	2478
1.855	2366	2389	2370	2.005	2474	2501	2482
1.860	2370	2393	2374	2.010	2478	2505	2485
1.865	2373	2397	2378	2.015	2482	2508	2489
1.870	2377	2400	2382	2.020	2485	2512	2492
1.875	2381	2404	2385	2.025	2489	2515	2496

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			Thickness, inches	Required BL(P), fps		
	Class I	Class II	Class III		Class I	Class II	Class III
2.030	2492	2519	2500	2.205	2612	2643	2623
2.035	2496	2523	2503	2.210	2615	2646	2626
2.040	2499	2526	2507	2.215	2619	2650	2629
2.045	2503	2530	2510	2.220	2622	2653	2633
2.050	2506	2534	2514	2.225	2625	2657	2636
2.055	2510	2537	2518	2.230	2629	2660	2640
2.060	2513	2541	2521	2.235	2632	2664	2643
2.065	2517	2544	2525	2.240	2635	2667	2647
2.070	2520	2548	2528	2.245	2639	2671	2650
2.075	2523	2552	2532	2.250 [3/]	2642	2674	2653
2.080	2527	2555	2535	2.255	2645	2678	2657
2.085	2530	2559	2539	2.260	2649	2681	2660
2.090	2534	2562	2542	2.265	2652	2685	2663
2.095	2537	2566	2546	2.270	2655	2688	2667
2.100	2541	2569	2550	2.275	2659	2694	2670
2.105	2544	2573	2553	2.280	2662	---	2674
2.110	2548	2576	2557	2.285	2665	---	2677
2.115	2551	2580	2560	2.290	2668	---	2680
2.120	2554	2584	2564	2.295	2672	---	2684
2.125	2558	2587	2567	2.300	2675	---	2687
2.130	2561	2591	2571	2.305	2678	---	2690
2.135	2565	2594	2574	2.310	2681	---	2694
2.140	2568	2598	2578	2.315	2685	---	2697
2.145	2572	2601	2581	2.320	2688	---	2700
2.150	2575	2605	2585	2.325	2691	---	2704
2.180	2595	2626	2605	2.330	2694	---	2707
2.185	2599	2629	2609	2.335	2698	---	2710
2.190	2602	2633	2612	2.340	2701	---	2714
2.195	2605	2636	2616	2.345	2704	---	2717
2.200	2609	2640	2619	2.350	2707	---	2720

[3] Specification class II requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
1.395	1997	1907	2007	1990
1.400	2001	1911	2011	1995
1.405	2006	1916	2016	1999
1.410	2010	1920	2020	2004
1.415	2015	1925	2025	2008
1.420	2019	1929	2029	2013
1.425	2023	1933	2034	2017
1.430	2028	1938	2038	2022
1.435	2032	1942	2043	2026
1.440	2036	1946	2047	2031
1.445	2040	1950	2052	2035
1.450	2045	1955	2056	2040
1.455	2049	1959	2061	2044
1.460	2053	1963	2065	2048
1.465	2058	1968	2069	2053
1.470	2062	1972	2074	2057
1.475	2066	1976	2078	2062
1.480	2070	1980	2083	2066
1.485	2074	1984	2087	2070
1.490	2079	1989	2092	2075
1.495	2083	1993	2096	2079
1.500	2087	1997	2100	2083
1.501 1/, 2/	2088	1998	2101	2084
1.505	2091	2001	2105	2088
1.510	2095	2005	2109	2092
1.515	2100	2010	2113	2096
1.520	2104	2014	2118	2100
1.525	2108	2018	2122	2105
1.530	2112	2022	2126	2109
1.535	2116	2026	2131	2113
1.540	2120	2030	2135	2118
1.545	2124	2034	2139	2122
1.550	2129	2039	2143	2126

1/ Specification requirements begin for this ordered thickness.
2/ Specification class I, type B minimum required ballistic limits shall be -90fps vs. listed thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
1.555	2133	2043	2148	2130
1.560	2137	2047	2152	2134
1.565	2141	2051	2156	2139
1.570	2145	2055	2160	2143
1.575	2149	2059	2165	2147
1.580	2153	2063	2169	2151
1.585	2157	2067	2173	2155
1.590	2161	2071	2177	2160
1.595	2165	2075	2181	2164
1.600	2169	2079	2185	2168
1.605	2173	2083	2190	2172
1.610	2177	2087	2194	2176
1.615	2181	2091	2198	2180
1.620	2185	2095	2202	2184
1.625	2189	2099	2206	2189
1.630	2193	2103	2210	2193
1.635	2197	2107	2214	2197
1.640	2201	2111	2219	2201
1.645	2205	2115	2223	2205
1.650	2209	2119	2227	2209
1.655	2213	2123	2231	2213
1.660	2217	2127	2235	2217
1.665	2221	2131	2239	2221
1.670	2225	2135	2243	2225
1.675	2229	2139	2247	2229
1.680	2233	2143	2251	2233
1.685	2237	2147	2255	2237
1.690	2241	2151	2259	2241
1.695	2245	2155	2263	2245
1.700	2248	2158	2267	2249
1.705	2252	2162	2271	2253
1.710	2256	2166	2275	2257
1.715	2260	2170	2279	2261

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
1.720	2264	2174	2283	2265
1.725	2268	2178	2287	2269
1.730	2272	2182	2291	2273
1.735	2276	2186	2295	2277
1.740	2279	2189	2299	2281
1.745	2283	2193	2303	2285
1.750	2287	2197	2307	2289
1.755	2291	2201	2311	2293
1.760	2295	2205	2315	2297
1.765	2298	2208	2319	2301
1.770	2302	2212	2323	2305
1.775	2306	2216	2327	2309
1.780	2310	2220	2331	2313
1.785	2314	2224	2335	2316
1.790	2317	2227	2339	2320
1.795	2321	2231	2343	2324
1.800	2325	2235	2346	2328
1.805	2329	2239	2350	2332
1.810	2332	2242	2354	2336
1.815	2336	2246	2358	2340
1.820	2340	2250	2362	2344
1.825	2344	2254	2366	2347
1.830	2347	2257	2370	2351
1.835	2351	2261	2374	2355
1.840	2355	2265	2377	2359
1.845	2359	2269	2381	2363
1.850	2362	2272	2385	2366
1.855	2366	2276	2389	2370
1.885	2388	2298	2412	2393
1.890	2392	2302	2415	2397
1.895	2395	2305	2419	2400
1.900	2399	2309	2423	2404
1.905	2403	2313	2427	2408

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
1.910	2406	2316	2431	2412
1.915	2410	2320	2434	2415
1.920	2414	2324	2438	2419
1.925	2417	2327	2442	2423
1.930	2421	2331	2446	2427
1.935	2424	2334	2449	2430
1.940	2428	2338	2453	2434
1.945	2432	2342	2457	2438
1.950	2435	2345	2460	2441
1.955	2439	2349	2464	2445
1.960	2442	2352	2468	2449
1.965	2446	2356	2472	2452
1.970	2450	2360	2475	2456
1.975	2453	2363	2479	2460
1.980	2457	2367	2483	2463
1.985	2460	2370	2486	2467
1.990	2464	2374	2490	2471
1.995	2467	2377	2494	2474
2.000	2471	2381	2497	2478
2.005	2474	2384	2501	2482
2.010	2478	2388	2505	2485
2.015	2482	2392	2508	2489
2.020	2485	2395	2512	2492
2.025	2489	2399	2515	2496
2.030	2492	2402	2519	2500
2.035	2496	2406	2523	2503
2.040	2499	2409	2526	2507
2.045	2503	2413	2530	2510
2.050	2506	2416	2534	2514
2.055	2510	2420	2537	2518
2.060	2513	2423	2541	2521
2.065	2517	2427	2544	2525
2.070	2520	2430	2548	2528

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
2.075	2523	2433	2552	2532
2.080	2527	2437	2555	2535
2.085	2530	2440	2559	2539
2.090	2534	2444	2562	2542
2.095	2537	2447	2566	2546
2.100	2541	2451	2569	2550
2.105	2544	2454	2573	2553
2.110	2548	2458	2576	2557
2.115	2551	2461	2580	2560
2.120	2554	2464	2584	2564
2.125	2558	2468	2587	2567
2.130	2561	2471	2591	2571
2.135	2565	2475	2594	2574
2.140	2568	2478	2598	2578
2.145	2572	2482	2601	2581
2.150	2575	2485	2605	2585
2.155	2578	2488	2608	2588
2.160	2582	2492	2612	2592
2.165	2585	2495	2615	2595
2.170	2588	2498	2619	2598
2.175	2592	2502	2622	2602
2.180	2595	2505	2626	2605
2.185	2599	2509	2629	2609
2.190	2602	2512	2633	2612
2.195	2605	2515	2636	2616
2.200	2609	2519	2640	2619
2.205	2612	2522	2643	2623
2.210	2615	2525	2646	2626
2.215	2619	2529	2650	2629
2.220	2622	2532	2653	2633
2.225	2625	2535	2657	2636
2.230	2629	2539	2660	2640
2.235	2632	2542	2664	2643

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
2.240	2635	2545	2667	2647
2.245	2639	2549	2671	2650
2.250 3/	2642	2552	2674	2653
2.255	2645	2555	2678	2657
2.260	2649	2559	2681	2660
2.265	2652	2562	2685	2663
2.270	2655	2565	2688	2667
2.275	2659	2569	2694	2670
2.280	2662	2572	2694	2674
2.285	2665	2575	2698	2677
2.290	2668	2578	2701	2680
2.295	2672	2582	2705	2684
2.300	2675	2585	2708	2687
2.305	2678	2588	2711	2690
2.310	2681	2591	2715	2694
2.315	2685	2595	2718	2697
2.320	2688	2598	2721	2700
2.325	2691	2601	2725	2704
2.330	2694	2604	2728	2707
2.335	2698	2608	2731	2710
2.340	2701	2611	2735	2714
2.345	2704	2614	2738	2717
2.350	2707	2617	2741	2720
2.355	2711	2621	2745	2724
2.360	2714	2624	2748	2727
2.365	2717	2627	2751	2730
2.370	2720	2630	---	2733
2.375	2724	2634	---	2737
2.380	2727	2637	---	2740
2.385	2730	2640	---	2743
2.390	2733	2643	---	2747
2.395	2736	2646	---	2750
2.400	2740	2650	---	2753

3/ Specification class II requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
2.405	2743	2653	---	2756
2.410	2746	2656	---	2760
2.415	2749	2659	---	2763
2.420	2752	2662	---	2766
2.425	2755	2665	---	2769
2.430	2759	2669	---	2773
2.435	2762	2672	---	2776
2.440	2765	2675	---	2779
2.445	2768	2678	---	2782
2.450	2771	2681	---	2786
2.455	2774	2684	---	2789
2.460	2778	2688	---	2792
2.465	2781	2691	---	2795
2.470	2784	2694	---	2799
2.475	2787	2697	---	2802
2.480	2790	2700	---	2805
2.485	2793	2703	---	2808
2.490	2796	2706	---	2811
2.495	2800	2710	---	2815
2.500 4/	2803	2713	---	2818
2.505	2806	2716	---	2821
2.510	2809	2719	---	2824
2.515	2812	2722	---	2827
2.520	2815	2725	---	2831
2.525	2818	2728	---	2834
2.530	2821	2731	---	2837
2.535	2824	2734	---	2840
2.540	2828	2738	---	2843
2.545	2831	2741	---	2846
2.550	2834	2744	---	2850
2.555	2837	2747	---	2853
2.560	2840	2750	---	2856
2.565	2843	2753	---	2859

4/ Specification class I, type A, and class III requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps			
	Class I, Type A	Class I, Type B	Class II	Class III
2.570	2846	2756	---	2862
2.575	2849	2759	---	2865
2.580	2852	2762	---	2868
2.585	2855	2765	---	2872
2.590	2858	2768	---	2875
2.595	2861	2771	---	2878
2.600 5/	2864	2774	---	2881
2.605	2867	2777	---	2884
2.610	2871	2781	---	2887
2.615	2874	2784	---	2890
2.620	2877	2787	---	2893
2.625	2880	2790	---	2897
2.630	2883	2793	---	---
2.635	2886	2796	---	---
2.640	2889	2799	---	---
2.645	2892	2802	---	---
2.650	2895	2805	---	---
2.655	2898	2808	---	---
2.660	2901	2811	---	---
2.665	2904	2814	---	---
2.670	2907	2817	---	---
2.675	2910	2820	---	---
2.680	2913	2823	---	---
2.685	2916	2826	---	---
2.690	2919	2829	---	---
2.695	2922	2832	---	---
2.700	2925	2835	---	---
2.705	2928	2838	---	---
2.710	2931	2841	---	---
2.715	2934	2844	---	---
2.720	2937	2847	---	---
2.725	2940	2850	---	---

5/ Specification class I, type B requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I		Class I
2.375	2507	2.530	2609
2.380	2511	2.535	2613
2.385	2514	2.540	2616
2.390	2517	2.545	2619
2.395	2521	2.550	2622
2.400	2524	2.555	2625
2.405	2527	2.560	2629
2.410	2531	2.565	2632
2.415	2534	2.570	2635
2.420	2537	2.575	2638
2.425	2541	2.580	2641
2.430	2544	2.585	2645
2.435	2547	2.590	2648
2.440	2551	2.595	2651
2.445	2554	2.600	2654
2.450	2557	2.605	2657
2.455	2561	2.610	2661
2.460	2564	2.615	2664
2.465	2567	2.620	2667
2.470	2571	2.625	2670
2.475	2574	2.630	2673
2.480	2577	2.635	2676
2.485	2580	2.640	2680
2.490	2583	2.645	2683
2.495	2586	2.650	2686
2.500	2590	2.655	2689
2.501 1/	2591	2.660	2692
2.505	2593	2.665	2695
2.510	2596	2.670	2698
2.515	2600	2.675	2702
2.520	2603	2.680	2705
2.525	2606	2.685	2708

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I		Class I
2.690	2711	2.850	2809
2.695	2714	2.855	2812
2.700	2717	2.860	2815
2.705	2720	2.865	2818
2.710	2723	2.870	2821
2.715	2727	2.875	2824
2.720	2730	2.880	2827
2.725	2733	2.885	2830
2.730	2736	2.890	2833
2.735	2739	2.895	2836
2.740	2742	2.900	2839
2.745	2745	2.905	2842
2.750	2748	2.910	2845
2.755	2751	2.915	2848
2.760	2754	2.920	2851
2.765	2757	2.925	2854
2.770	2760	2.930	2857
2.775	2763	2.935	2860
2.780	2767	2.940	2863
2.785	2770	2.945	2866
2.790	2773	2.950	2869
2.795	2776	2.955	2872
2.800	2779	2.960	2874
2.805	2782	2.965	2877
2.810	2785	2.970	2880
2.815	2788	2.975	2883
2.820	2791	2.980	2886
2.825	2794	2.985	2889
2.830	2797	2.990	2892
2.835	2800	2.995	2895
2.840	2803	3.000	2898
2.845	2806	3.005	2901

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I		Class I
3.010	2904	3.165	2993
3.015	2907	3.170	2995
3.020	2910	3.175	2998
3.025	2912	3.180	3001
3.030	2915	3.185	3004
3.035	2918	3.190	3007
3.040	2921	3.195	3010
3.045	2924	3.200	3012
3.050	2927	3.205	3015
3.055	2930	3.210	3018
3.060	2933	3.215	3021
3.065	2936	3.220	3024
3.070	2938	3.225	3026
3.075	2941	3.230	3029
3.080	2944	3.235	3032
3.085	2947	3.240	3035
3.090	2950	3.245	3037
3.095	2953	3.250	3040
3.100	2956	3.255	3043
3.105	2959	3.260	3046
3.110	2961	3.265	3049
3.115	2964	3.270	3051
3.120	2967	3.275	3054
3.125	2970	3.280	3057
3.130	2973	3.285	3060
3.135	2976	3.290	3062
3.140	2978	3.295	3065
3.145	2981	3.300	3068
3.150	2984	3.305	3071
3.155	2987	3.310	3073
3.160	2990	3.315	3076

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I		Class I
3.320	3079	3.475	3163
3.325	3082	3.480	3166
3.330	3084	3.485	3168
3.335	3087	3.490	3171
3.340	3090	3.495	3174
3.345	3093	3.500 2/	3176
3.350	3095	3.505	3179
3.355	3098	3.510	3182
3.360	3101	3.515	3184
3.365	3104	3.520	3187
3.370	3106	3.525	3190
3.375	3109	3.530	3193
3.380	3112	3.535	3195
3.385	3114	3.540	3198
3.390	3117	3.545	3201
3.395	3120	3.550	3203
3.400	3123	3.555	3206
3.405	3125	3.560	3209
3.410	3128	3.565	3211
3.415	3131	3.570	3214
3.420	3133	3.575	3216
3.425	3136	3.580	3219
3.430	3139	3.585	3222
3.435	3141	3.590	3224
3.440	3144	3.595	3227
3.445	3147	3.600	3230
3.450	3150	3.605	3232
3.455	3152	3.610	3235
3.460	3155	3.615	3237
3.465	3158	3.620	3240
3.470	3160	3.625	3243

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VII. Minimum required ballistic limits - 20mm M602 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I		Class I
3.480	TBD	3.635	TBD
3.485	TBD	3.640	TBD
3.490	TBD	3.645	TBD
3.495	TBD	3.650	TBD
3.501 1/	TBD	3.655	TBD
3.505	TBD	3.660	TBD
3.510	TBD	3.665	TBD
3.515	TBD	3.670	TBD
3.520	TBD	3.675	TBD
3.525	TBD	3.680	TBD
3.530	TBD	3.685	TBD
3.535	TBD	3.690	TBD
3.540	TBD	3.695	TBD
3.545	TBD	3.700	TBD
3.550	TBD	3.705	TBD
3.555	TBD	3.710	TBD
3.560	TBD	3.715	TBD
3.565	TBD	3.720	TBD
3.570	TBD	3.725	TBD
3.575	TBD	3.730	TBD
3.580	TBD	3.735	TBD
3.585	TBD	3.740	TBD
3.590	TBD	3.745	TBD
3.595	TBD	3.750	TBD
3.600	TBD	3.755	TBD
3.605	TBD	3.760	TBD
3.610	TBD	3.765	TBD
3.615	TBD	3.770	TBD
3.620	TBD	3.775	TBD
3.625	TBD	3.780	TBD
3.630	TBD	3.785	TBD

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VII. Minimum required ballistic limits - 20mm M602 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
	Class I		Class I
3.790	TBD	3.915	TBD
3.795	TBD	3.920	TBD
3.800	TBD	3.925	TBD
3.805	TBD	3.930	TBD
3.810	TBD	3.935	TBD
3.815	TBD	3.940	TBD
3.820	TBD	3.945	TBD
3.825	TBD	3.950	TBD
3.830	TBD	3.955	TBD
3.835	TBD	3.960	TBD
3.840	TBD	3.965	TBD
3.845	TBD	3.970	TBD
3.850	TBD	3.975	TBD
3.855	TBD	3.980	TBD
3.860	TBD	3.985	TBD
3.865	TBD	3.990	TBD
3.870	TBD	3.995	TBD
3.875	TBD	4.000 2/	TBD
3.880	TBD	4.010	TBD
3.885	TBD	4.015	TBD
3.890	TBD	4.020	TBD
3.895	TBD	4.025	TBD
3.900	TBD	4.030	TBD
3.905	TBD	4.035	TBD
3.910	TBD	---	---

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY - MR

(Project 9535-2025-004)

Review activities:
Army - AR, AT, AV, TE
DLA - CQ

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